Ex:-4:- 60% of 50% of 40% of 30% of 20% of 10% of 60,000?

Ans- M1:- Basic:- $\frac{60}{100} \times \frac{50}{100} \times \frac{40}{100} \times \frac{30}{100} \times \frac{20}{100} \times \frac{10}{100} \times 60000$

$= \frac{4320}{100} = 43.2$

M2:- By using logic of % assuming Initial as 100%.

60% of	50% of	40% of	30% of	20% of	10% of 60000
					→ 6000
				→ 1200	
			→ 360		
		→ 144			
	→ 72				
(43.2)					

Ex:-5:- In Amazon-Prime day sale a watch of worth ₹. 25,000/- Rs decreased by 20%, due to this much discount demand ↑ed, then again ↑ed by 30% on discounted price. the net % change and final selling price Rs?

Ans- M1:-

25,000
- 5,000 -20% of 25k ↓ed
20,000
6000 -30% of 20k ↑ed
26,000 → final Selling Price

Net % change = $\frac{\text{final} - \text{Initial}}{\text{Initial}} \times 100$

$= \frac{26000 - 25000}{25000} \times 100$

$= 4\%$ ↑ed

M2:- By using decimal factor multiplication.

Final SP = $1.3 \times 0.8 \times 25000$

$= 1.04 \times 25000$

$= 1.04 \times 100 \times 25000$

$= 104\%$ of 25000

F.SP = 26000

$= (100+4)\%$ of 25000

Net change = 4% ↑ed

M3:- By using formula:- $\left[\pm x \pm y \pm \frac{xy}{100}\right]\%$

here first $x = 20\%$ ↓ed, next $y = 30\%$ ↑ed $[-\ +]$

Net % change $= -20 + 30 - \frac{20 \times 30}{100}\% = 10 - \frac{600}{100}\% = 4\%$

Final Selling Price $= (100+4)\%$ of 25000

$= 104\%$ of 25000

$= 25000 + 4\%$ of 25000

$= 25000 + 1000$

$= 26000$

Ex:6:- Sunil saves 20% of his income generally. If salary ↑ed by 30%, then he ↑ed his expenditure by 20%. The % change in his savings is ?

Ans:- Assume Initial salary = 100%

	Salary	Expenditure	Savings
Initial	100%	80%	20%
Final	100% + 30% = 130%	80% + 30% of 80% = 80% + 24% = 104%	26%

Note:- Dont consider direct diff as % change.

% change in savings

$$= \frac{F-I}{I} \times 100 = \frac{(26-20)\%}{20\%} \times 100$$

= 30% ↑ed

(or)

assume 20% = Initial Savings = 100% of 20%

2% = 10% of 20%

Changed to 26% means diff is 6%

i.e 2% is 10% of 20%

then 6% is 30% of 20%. So, 30% ↑ed in savings.

Ex:7:- Satish spending his salary every month in such a way that, 20% of his income on groceries, 30% of the remaining on rent, 10% of the remaining on food and 10% of the income on entertainment. Finally he left out with 8080/- as savings. The Initial income / salary is ?

Ans:- 100% = Initial / Income

20%, 100% − 20% = 80%

30% of 80% = 24%, 80% − 24% = 56%

10% of 56% = 5.6%, 56 − 5.6% = 50.4%

50.4% − 10% = 40.4% = Rem

40.4% = 8080

1% = $\frac{8080}{40.4}$ = 200

100% = 20,000

(or) decimal method:- Total salary as 'x'

$$0.9 \times 0.7 \times 0.8x - 0.1x = 8080$$

$$0.404x = 8080$$

$$x = 20000$$

Excercise Questions:- [Booklet Questions]

Ex:8:-[B.Q:-5]:- The monthly income of shyam & kamal together Rs 28000. The income of shyam an kamal was fed by 25% and 12.5% respectively. The new income of kamal becomes 120% of new salary of shyam. What is the new income of shyam?

Ans:- $S + K = 28000$, —①

S_n

New incomes $\Rightarrow 1.25S$, $1.125K = K_n$

K_n = 120% of $S_n \Rightarrow 1.125k = 1.2 \times 1.25S = 1.5S$

$\Rightarrow K = \frac{1.5}{1.125}S = \frac{150}{112.5}S = \frac{4}{3}S$ —②

Sub ② in ① $\Rightarrow S + \frac{4}{3}S = 28000 \Rightarrow \frac{7}{3}S = 28000$

$\Rightarrow S = 12000$ But

We want $S_n = 1.25S = \frac{5}{4}S = 15000/-$

Ex:9:- [B.Q:-6]:- The max marks in 3 subjects M, P & C are in 1:2:3 respectively. Giri obtained 40% in M, 60% in P & 35% in C. What is overall % giri get?

Ans:- $M : P : C = 1 : 2 : 3 = x : 2x : 3x$

$$\% \text{ of Marks} = \frac{\text{Obtained}}{\text{Total}} \times 100 = \frac{40\% \text{ of } x + 60\% \text{ of } 2x + 35\% \text{ of } 3x}{x + 2x + 3x} \times 100$$

$$= \frac{0.4x + 0.6 \times 2x + 0.35 \times 3x}{6x} \times 100 = \frac{2.65x}{6x} \times 100$$

$$= \frac{265}{6}\% = 44\frac{1}{6}\% = 44.166\% \cong 44\%$$

Ex:-10:-[B.Q:7]:- Fresh fruit contains 75% of water, while dry fruit contains 20% of water. If the weight of dry fruit is 300 kgs. What was it total weight when it was fresh?

Ans:- Fresh Fruit → 75% of water ⇒ 25% of Pulp

Dry Fruit → 20% of water ⇒ 80% of Pulp

Given 300 kgs D_{fruit} ⇒ 20% = 60kgs water & 80% = 240kgs pulp

If the above case with fresh fruit means pulp is to be considered as 25%

⇒ 25% of pulp = 240kgs = $\frac{1}{4}$ of fresh fruit

100% of fresh fruit = $\frac{1}{4} \times 4 = 240 \times 4 =$ **960kgs.**

Ex:-11:-[B.Q:8]:- The total sal of Pandu & Harish is Rs 30000. If Pandu sal ↑ed 5% & harish ↑ed by 7%. Then their total salary 31800%. Find harish sal?

$[P + H = 30000] 53$ ①

$[1.05P + 1.07H = 31800] 50$

$53P + 53H = 52.5P + 53.5H$

$\Rightarrow 0.5P = 0.5H$

$\Rightarrow P = H$ — ②

Sub ② in ① $H + H = 30000$

$\Rightarrow$ **H = 15000**

300	30000, 31800
2	100, 106
	50, 53

$300 \times 2 \times 50 \times 53 = 1590000$

$= 30000 \times 53$

$= 31800 \times 50$

Ex:-12:-[B.Q:11]:- Deepika went to a fruit shop with a certain amount of money. She retains 15% of it to auto, with remaining she can buy 40 Apples or 70 Oranges. If she buy 35 Oranges, How many more Apples she can buy?

Ans:- 100% − 15% = 85% = 40 A = 70 Or ⇒ [20A + 20A] = [350 + 350]

She bought 35 oranges means 20 A out of 70 Oran = 40 A.

So, she can still buy **20 Apples**